

the stimulus and thereby shutting off secretin release (see Figure 45.9). By decreasing hormone signaling, negative-feedback regulation prevents excessive pathway activity.

Whereas negative feedback dampens a stimulus, **positive feedback** reinforces a stimulus, driving a process to completion. For example, in the pathway outlined in Figure 45.10, the mammary glands secrete milk in response to circulating oxytocin. The released milk in turn leads to more suckling and thus more stimulation. Activation of the pathway is sustained until the baby is full and stops suckling. Other functions of oxytocin, such as stimulating contractions of the uterus during birthing, also exhibit positive feedback.

Comparing negative and positive feedback, we see that only negative feedback helps restore a preexisting state. It is not surprising, therefore, that hormone pathways involved in homeostasis typically exhibit negative feedback. Often such pathways are paired, providing even more balanced control. For example, the blood glucose level is tightly regulated by the opposing, or antagonistic, effects of insulin and glucagon (see Figure 41.23).

Coordination of the Endocrine and Nervous Systems

In a wide range of animals, endocrine organs in the brain integrate function of the endocrine system with that of the

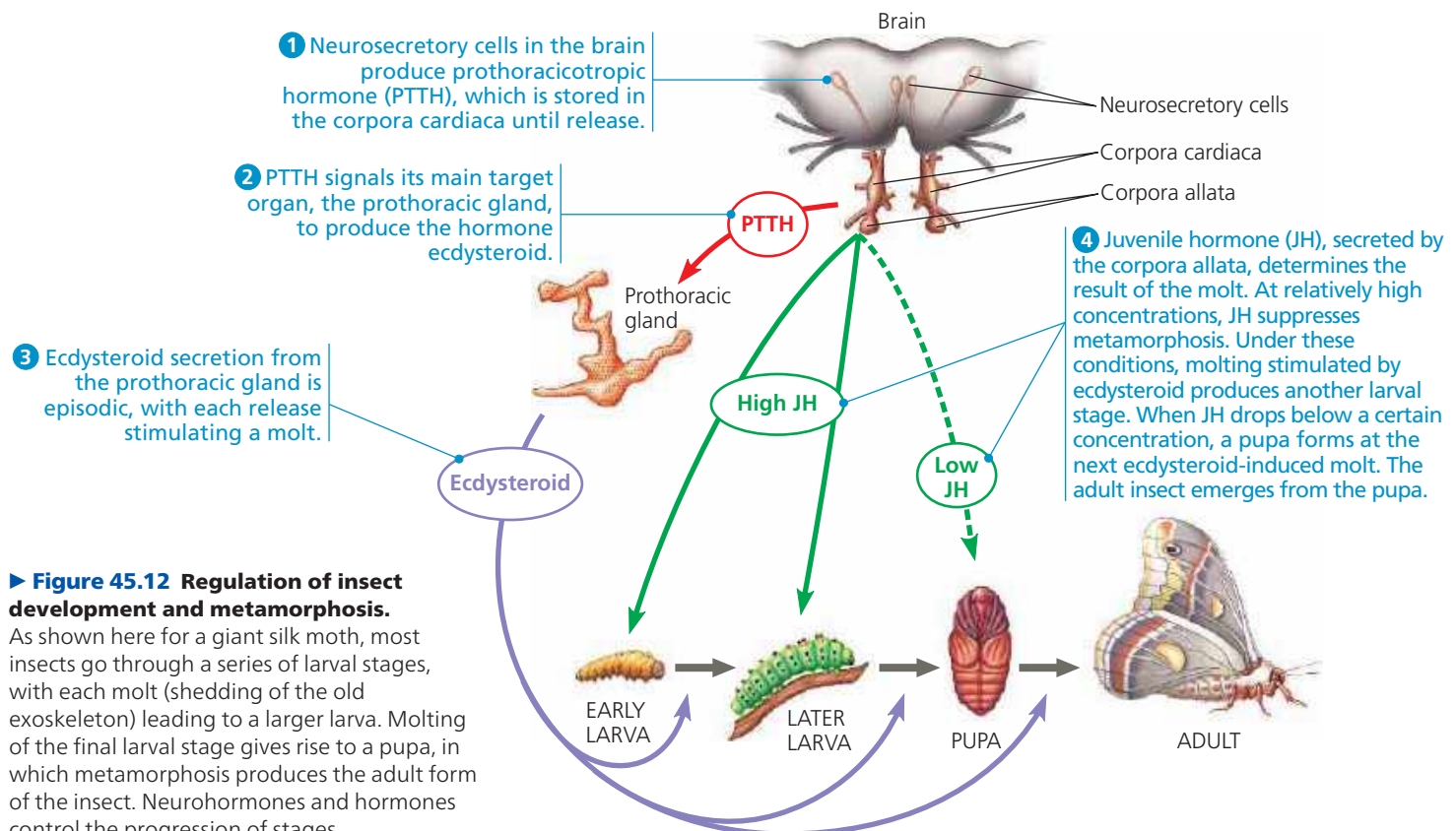
nervous system. We'll explore the basic principles of such integration in invertebrates and vertebrates.

Invertebrates

The control of development in a moth illustrates neuroendocrine coordination in invertebrates. A moth larva, such as the caterpillar of the giant silk moth (*Hyalophora cecropia*) shown in **Figure 45.11**, grows in stages. Because its exoskeleton cannot stretch, the larva must periodically molt, shedding the old exoskeleton and secreting a new one. The endocrine pathway that controls molting originates in the larval brain (**Figure 45.12**). Neurosecretory cells in the brain produce PTTH, a polypeptide neurohormone. When PTTH in body fluids reaches an endocrine organ called the prothoracic gland, it directs release of a second hormone, *ecdysteroid*. Bursts of *ecdysteroid* trigger each successive molt.

Ecdysteroid also controls a remarkable change in form called metamorphosis. Within the larva lie islands of tissues that will become the eyes, wings, brain, and other adult structures. Once a plump, crawling larva becomes a stationary pupa, these islands of cells take over. They complete their program of development, while many larval tissues undergo programmed cell death. The end result is the transformation of the crawling caterpillar into a free-flying moth.

▼ **Figure 45.11** Larva of the giant silk moth.



► **Figure 45.12** Regulation of insect development and metamorphosis.

As shown here for a giant silk moth, most insects go through a series of larval stages, with each molt (shedding of the old exoskeleton) leading to a larger larva. Molting of the final larval stage gives rise to a pupa, in which metamorphosis produces the adult form of the insect. Neurohormones and hormones control the progression of stages.